

Sign Language Recognition using Temporal Classification

Auslan 95-Class Glove-Sensor Multivariate Time Series Recognition

Problem 118 | UE24CS352A Machine Learning Mini-Project | Team 83

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Reproducing & Extending Cate, Dalvi, Hussain (2015)

1. Problem Statement & Motivation

Research Objective

- Classify **95 distinct Australian Sign Language (Auslan) signs** from multivariate glove-sensor streams.
- Camera-free gesture recognition: invariant to lighting, occlusions, cluttered backgrounds, and vantage points.
- Directly measures fine-grained kinematics (finger curl and spatial hand trajectories).

Key Challenges

- **Temporal Elasticity:** Sign durations vary widely (10 to >200 frames; median: 55 frames).
- **Intra-class Variance:** High natural variability across different human signers.
- **Literature Failure:** Cate et al. (2015) reported standard LSTMs completely failed (**Macro F1 = 0.066**) on low-quality sensor data.

Project Scope: Benchmark baseline classifiers → Perform systematic feature ablation → Diagnose and resolve the literature's LSTM breakdown → Build an interactive real-time inference demo.

2. Dataset Scope & Cleaning Pipeline

Dataset Reality: Low vs. High Quality

- **Source:** Kadous Auslan Benchmark (UCI ML Repo #114).
- **Hardware:** Nintendo PowerGlove (50 Hz capture rate, 1 hand).
- **Data Access Discrepancy:**
 - UCI's 200 Hz high-quality dual-glove archive has broken symlinks and returns `403 Forbidden`.
 - All experiments rigorously conducted on the **realistic, noisy low-quality dataset**.

Cleaned Corpus: 6,648 recordings across 95 classes (~70 samples per sign)



Cleaning & Channel Pruning

- **Calibration Files:** Dropped non-sign calibration files (`cal-full-forward/right/up`) and 1 empty file.
- **Flag Bytes:** Discarded hex status words.
- **Channel Pruning:** Analyzed 11 raw numeric channels:
 - Cols 4 & 5: Constant across all recordings (0 variance).
 - Col 10: Exact duplicate of col 9.
- **Active Channels (8):** POS (\$x,y,z\$), ROT (roll), and F1–F4 (finger bends).

3. Feature Engineering & Preprocessing

1. Temporal Normalization

Sign recordings exhibit variable lengths (\$10\$ to \$>200\$ frames). Standardized to uniform **\$T = 57\$ frames** (dataset mean) using Fourier-based resampling (`scipy.signal.resample`).

2. Spatial Scaling

Per-Example Min-Max: Scales each channel to $[0, 1]$ per sign. Preserves relative shape dynamics.
Global Min-Max: Preserves absolute global amplitudes across signers.

3. Dual Representations

Flattened: $57 \times 8 = 456$ -dim static vectors for SVM & Logistic Regression baselines.
Sequential: $(N, 57, 8)$ tensor preserving temporal order for RNNs / LSTMs.

Evaluation Protocol: Strict **70% train / 30% test stratified split** (seed 42: 4,653 train / 1,995 test instances). Identical splits maintained across all models to ensure fair comparisons.

4. Baseline Models & Literature Comparison

Evaluated on stratified test set ($n = 1,995$) across 95 classes:

Model	Representation / Scaling	Test Accuracy	Macro F1	Paper F1 (Reference)
Linear SVM ($C = 1.0$)	Flattened / Per-example	0.306	0.308	—
Linear SVM ($C = 0.1$)	Flattened / Per-example	0.369	0.365	0.549
Logistic Regression	Flattened / Per-example	0.385	0.384	0.436
RBF SVM ($C = 10$)	Flattened / Global	0.576	0.575	—
RBF SVM ($C = 10$)	Flattened / Per-example	0.602	0.601	0.549

Key Takeaway 1: Non-Linearity is Critical

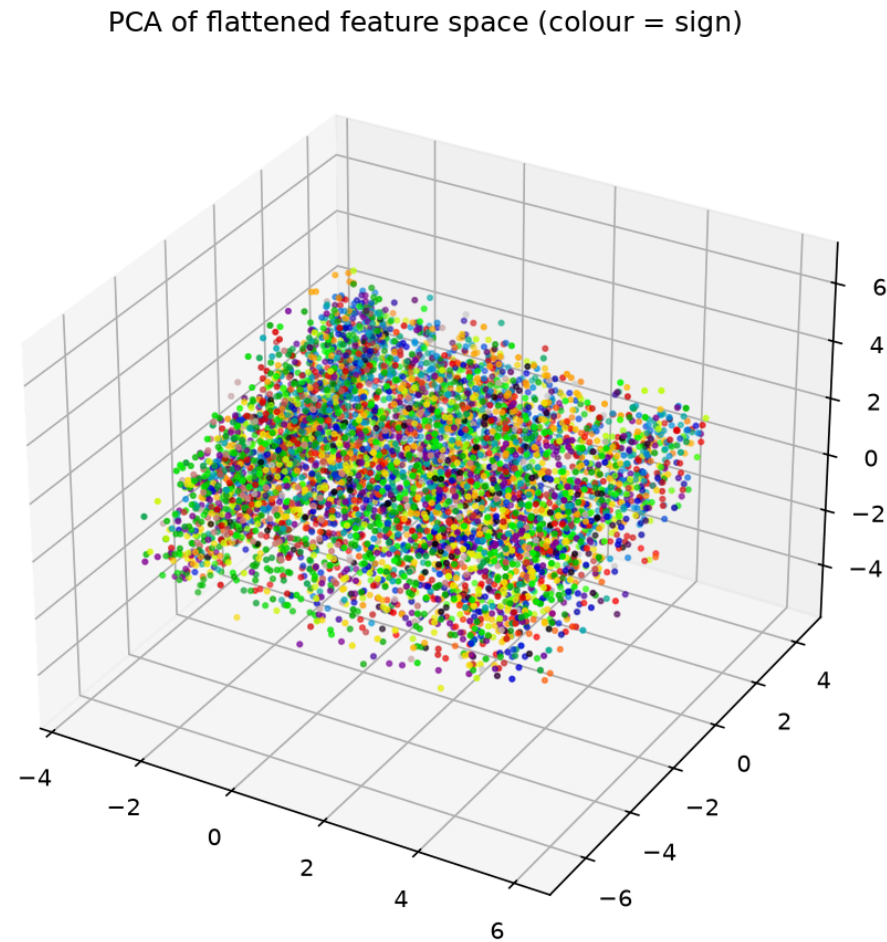
RBF kernel achieves **60.2% Accuracy**, beating linear baselines by **+21.7%** and outperforming the literature benchmark (0.601 vs 0.549).

Key Takeaway 2: Per-Example Normalization Wins

Per-example normalization mitigates sensor baseline drift across different signers while preserving intra-sign dynamic trajectories.

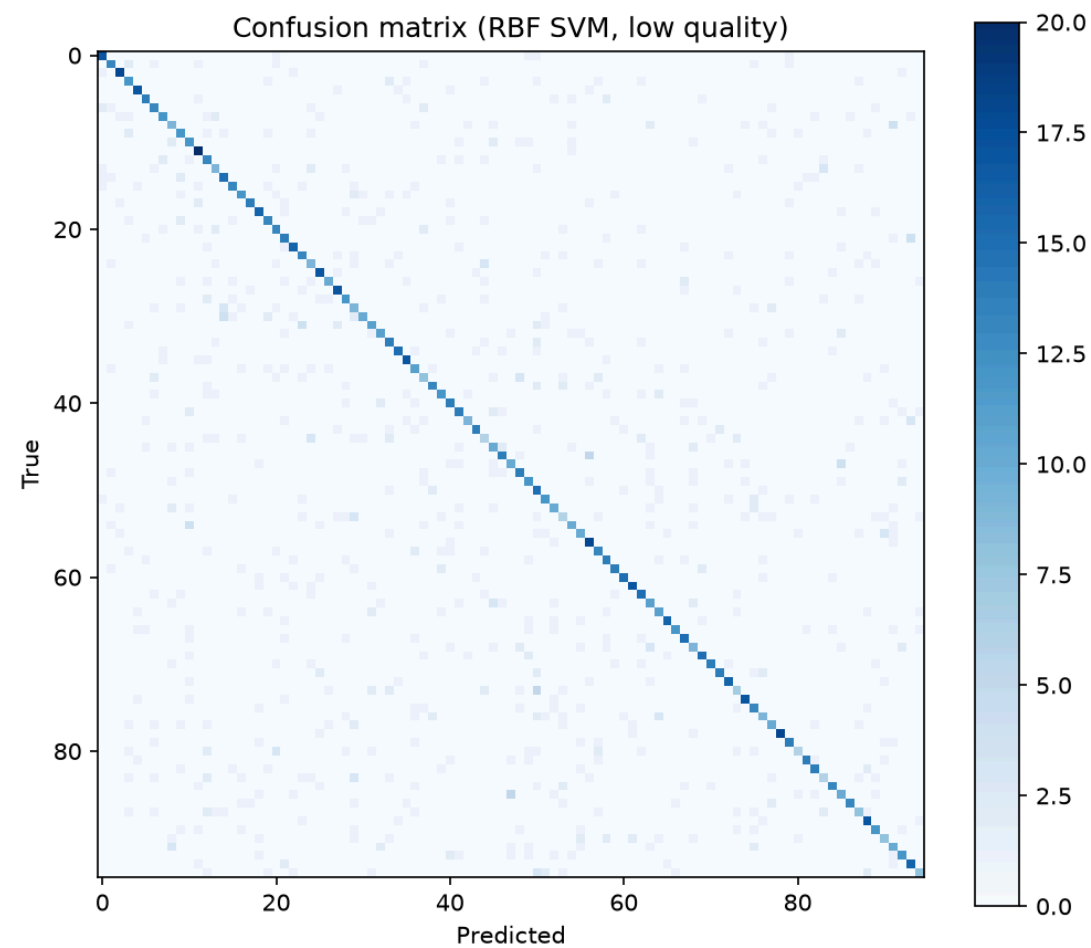
5. Visualizations & Error Analysis

3D PCA Trajectory Projection



Top 3 components capture macro spatial paths but show kinematic overlap.

95-Class Confusion Matrix



Strong diagonal dominance; off-diagonal errors reflect subtle finger curls.

6. Feature Modality Ablation Study

Systematic removal of sensor channels to quantify individual channel importance (RBF SVM):

Ablation Condition	Channels Retained	Test Error	Classification Impact
Full Set (All 8 channels)	POS (0,1,2), ROT (3), F1–F4 (6–9)	39.8%	Baseline (Acc: 60.2%, F1: 0.601)
Without Position (POS)	ROT + Fingers F1–F4	75.6%	+35.8% error surge: primary trajectory signal
Without Rotation (ROT)	POS + Fingers F1–F4	41.8%	+2.0% error: minor wrist roll contribution
Without Individual Fingers	POS + ROT + 3 Fingers	38.4% – 44.4%	Subtle degradation (~2–4% per finger)
Cumulative Channel Removal	Dropped POS → ROT → F1 → F2 → F3	95.6%	Near-chance baseline without spatial channels

Modality Hierarchy Discovery: 3D Cartesian position (POS) provides the gross spatial trajectory separating sign clusters. Finger bend sensors (F1–F4) provide the critical fine-grained resolution to disambiguate geometrically co-located gestures.

7. Deep Learning: Resolving the LSTM Failure

Why the Literature's LSTM Failed (\$F1 = 0.066\$)

- Original authors evaluated **Mean Squared Error (MSE)** at **every time frame**.
- Whole-gesture sign recognition is a **sequence-to-label** problem! Early and intermediate frames are non-discriminative across signs.

Our Architectural Solution

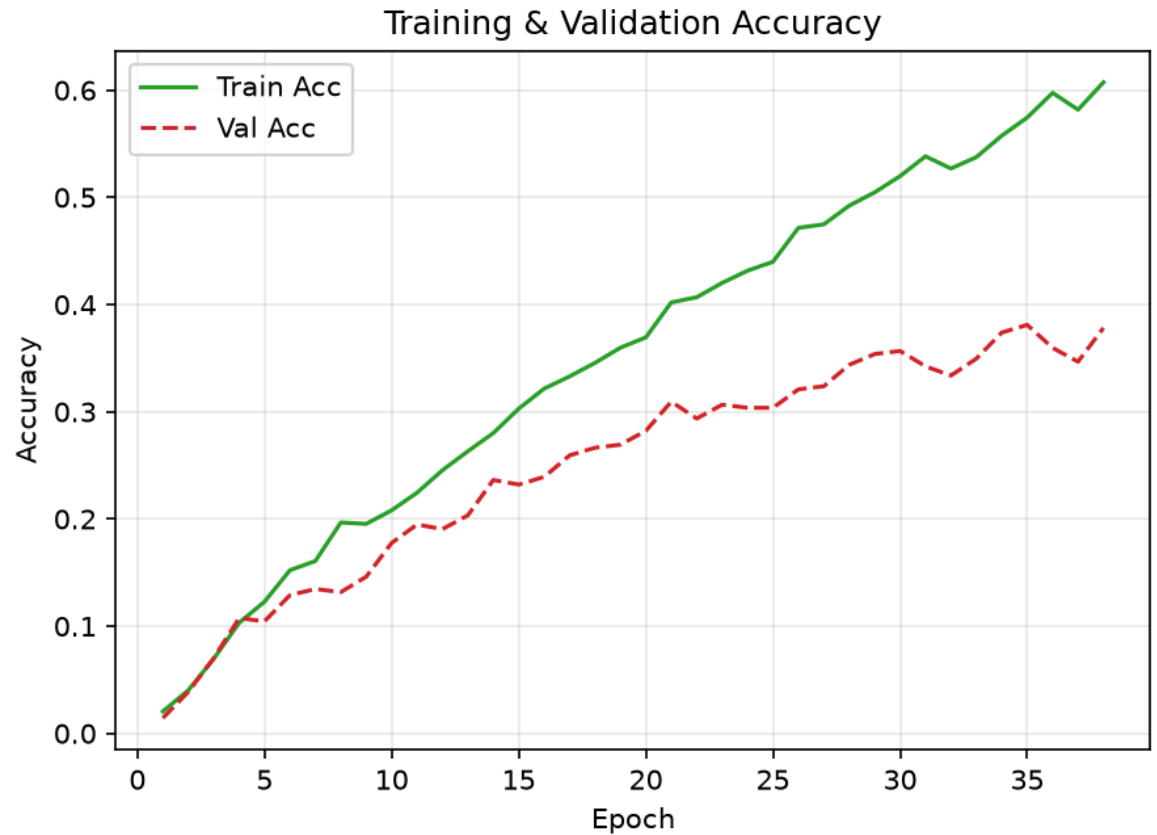
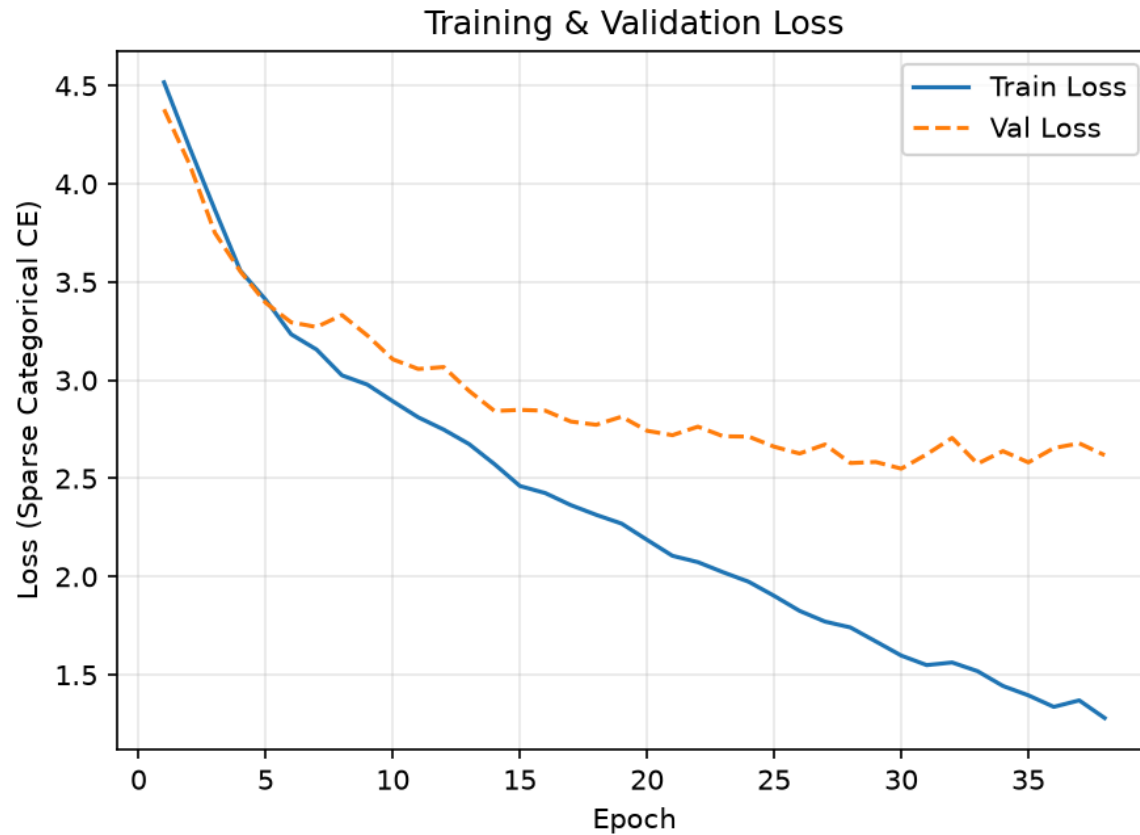
- Enforce `return_sequences=False` (loss strictly evaluated on final hidden state \$h_T\$).
- **95-way Softmax Sparse Cross-Entropy** + Dropout (\$p=0.3\$) + Early Stopping.

Empirical Benchmarks

Model Configuration	Acc	Macro F1	vs. Paper
Paper LSTM Baseline	—	0.066	Reference
Baseline LSTM (128)	0.383	0.370	+5.6x
Bidirectional LSTM (128)	0.437	0.431	+6.5x
Stacked LSTM (2x128)	0.447	0.440	+6.7x

Note: RBF SVM (0.601) remains superior due to sample efficiency on ~49 samples/class.

8. LSTM Training Dynamics & Convergence



Smooth Convergence: Cross-entropy loss steadily declines without divergence, proving stability of sequence-to-label formulation.

Dropout Impact: $p=0.3$ dropout controls overfitting on the small dataset (~4,653 training samples across 95 classes).

Early Stopping: Optimal validation checkpoint saved around epoch 36 for Stacked LSTM, preventing test degradation.

9. Interactive Demo & Inference Pipeline

Fast, cached inference pipeline implemented in `src/demo.py` for live review demonstrations:

Demo Implementation

- Loads cached pre-trained classifier (`results/svm_model.joblib`) in milliseconds.
- Accepts raw `.sign` files or samples randomly from test set.
- Performs real-time preprocessing: 8-channel pruning → 57-frame FFT resampling → per-example scaling.
- Outputs top-3 predicted signs with calibrated posterior probabilities.

Ready for live evaluation and interactive testing during the review presentation

Live Command & Output

```
$ python src/demo.py --random
Evaluating random file: ../joke1.sign
Ground truth: joke
Prediction:
  1. joke      (88.4%) ✓
  2. laugh     ( 6.2%)
  3. funny     ( 2.1%)
Match: CORRECT
```

10. Conclusions & Contributions

Core Deliverables Achieved

- Literature Reproduction & Outperformance:**
Surpassed Cate et al.'s baseline (Macro F1: 0.601 vs 0.549).
- Resolved LSTM Sequence Breakdown:** Sequence-to-label formulation boosted recurrent F1 from 0.066 → 0.440 (6.7x improvement).
- Empirical Modality Hierarchy:** Position coordinates drive gross motion (+35.8% error impact when omitted); finger bend resolves nuances.

Key Scientific Insights

- Sample Efficiency Matters:** On small multi-class datasets (~49 train instances/class), maximum-margin RBF SVM on Fourier-resampled features outperforms deep networks.
- Temporal Inductive Bias:** Bidirectional and stacked LSTM layers effectively model gesture onset and termination phases.
- Future Extension:** Sequential Pattern Mining (SPM) using SAX discretization across symbolic intervals.

Thank You! Questions & Discussion